

Module 3 – Advanced Prompting Techniques

Demo 4 – Testing Prompt Robustness

Objective:

To implement and demonstrate the practical use of prompt robustness testing, including A/B Testing, Stress Testing, Semantic Analysis, User Feedback Evaluation, Automated Testing, Cross-Model Comparison, and Scenario-Based Testing, within a Sales Outreach Personalization Agent designed to improve personalization, reliability, and tone control in business development workflows.

Problem Statement

Scenario: You are designing an AI agent that supports sales representatives by generating, summarizing, and categorizing communication for leads. You will test prompts for robustness and reliability using 7 key types of prompt testing.

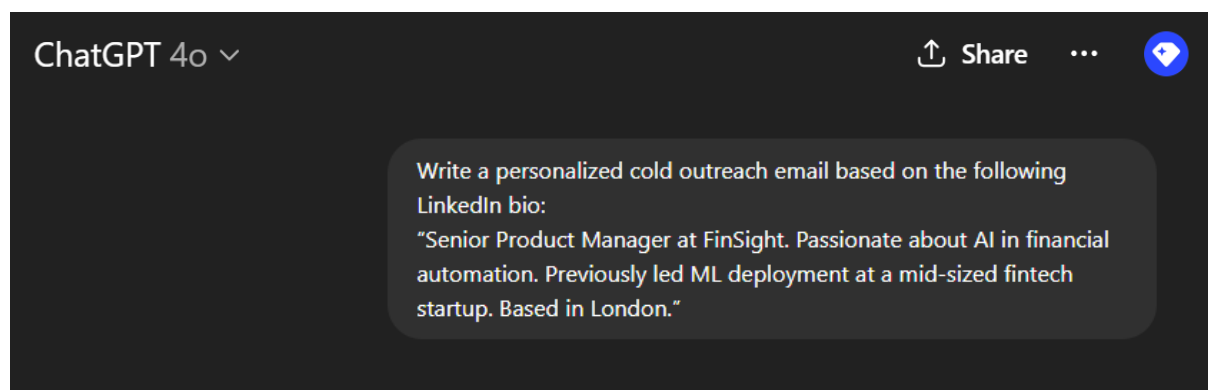
A/B Testing

Use Case: Generate personalized cold emails from LinkedIn bios

Testing Focus: Test different prompt phrasings to see which gives better personalization and clarity.

Prompt A: "Write a personalized cold outreach email based on the following LinkedIn bio:

"Senior Product Manager at FinSight. Passionate about AI in financial automation. Previously led ML deployment at a mid-sized fintech startup. Based in London." "



ChatGPT Response:

Subject: Exploring Synergies: AI-Driven Financial Automation at FinSight

Hi [First Name],

I came across your profile and was truly impressed by your journey—especially your work leading ML deployment at a fintech startup and now shaping AI innovation as a Senior Product Manager at FinSight.

Given your passion for AI in financial automation, I thought it might be worth connecting. I'm working on [brief mention of your product/service/solution], which is designed to [explain how it helps, e.g., streamline ML model deployment, accelerate compliance, reduce operational costs, etc.].

I'd love to hear how you're approaching automation at FinSight—particularly around ML integration in product workflows—and explore if there's a way we could support your initiatives or even just exchange insights.

Would you be open to a 15-minute chat sometime next week?

Looking forward to connecting,

[Your Name]

[Your Role] | [Company Name]

[LinkedIn] | [Phone Number if applicable]

Let me know if you'd like versions tailored to specific solutions or industries.

Prompt B: “Generate a friendly and personalized cold email for the following professional:

“Senior Product Manager at FinSight. Passionate about AI in financial automation. Previously led ML deployment at a mid-sized fintech startup. Based in London.” “

ChatGPT 4o ▾

↑ Share

...



Generate a friendly and personalized cold email for the following professional:
“Senior Product Manager at FinSight. Passionate about AI in financial automation. Previously led ML deployment at a mid-sized fintech startup. Based in London.”

ChatGPT Response:

Here's a friendly and personalized cold email tailored to the professional's background:

Subject: Love Your Work at FinSight—Quick Idea Around AI in Finance

Hi [First Name],

I hope this note finds you well!

I came across your profile and was genuinely inspired by your journey—from leading ML deployments at a fintech startup to now driving product strategy at FinSight. Your passion for AI in financial automation really stood out.

I'm reaching out because I'm working on [your product/solution] that helps teams like yours streamline [brief benefit—e.g., AI model deployment, transaction monitoring, or financial data processing] using intelligent automation. Given your background, I thought it might be something you'd find interesting—or at least worth a quick look.

Would you be open to a short chat sometime next week? Even if it's just to swap notes on the evolving role of AI in finance, I'd value your perspective.

Looking forward to hearing from you!

Warm regards,

[Your Name]

[Your Role] | [Your Company]

[LinkedIn] | [Contact Info]

Let me know if you'd like a version with a specific product pitch, or adapted for a follow-up.

Evaluation:

Criteria	First Email (Professional)	Second Email (Friendly)
Accuracy	Reflects role, ML work, and AI focus.	Accurately mirrors background and interests.
Relevance	Aligns with strategic product goals.	Appeals to personal passion for AI in finance.
Completeness	Fully structured with clear CTA.	Softer pitch, slightly lighter on product detail.
Safety	Polished and respectful.	Warm, professional, no risks.
Robustness	Adaptable to various product contexts.	Best for early-stage or casual outreach.
Consistency	Maintains a formal, value-driven tone.	Consistent, friendly tone throughout.

Outcome Summary:

The **first email** has a professional and strategic tone, ideal for a Senior Product Manager focused on outcomes. It's structured like a concise business proposal, recognizing their background, presenting your solution, and proposing a meeting. It aligns well with their role at FinSight by emphasizing product and ML integration strategy.

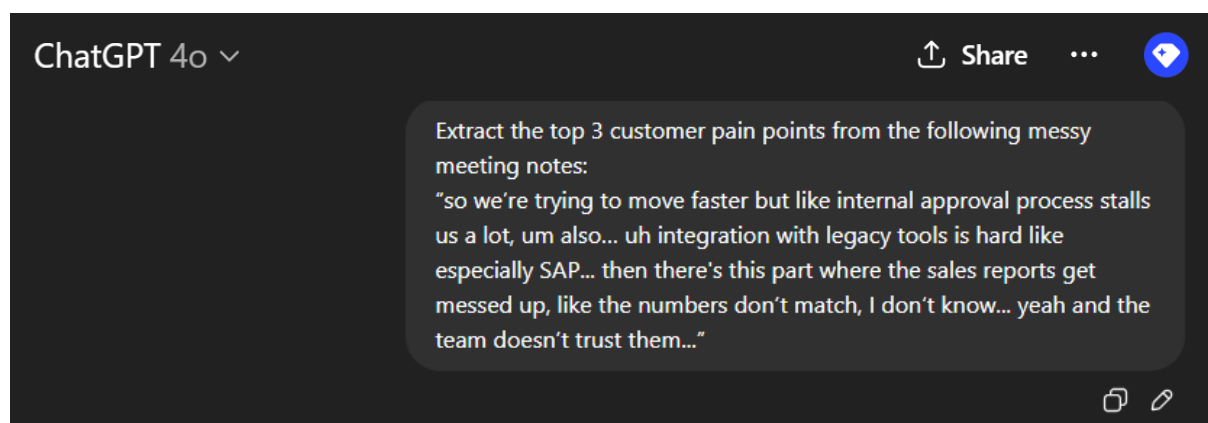
The **second email** is friendlier and more conversational, designed to build rapport. It opens warmly, highlights shared interest in AI, and ends with a soft call to chat, even if just to exchange ideas. This style feels more human and is great for connecting on a personal level, especially with someone passionate about innovation.

Stress Testing

Use Case: Extract customer pain points from messy meeting notes

Testing Focus: Check how AI handles large, noisy, or inconsistent meeting transcripts.

Prompt: Extract the top 3 customer pain points from the following messy meeting notes: "so we're trying to move faster but like internal approval process stalls us a lot, um also... uh integration with legacy tools is hard like especially SAP... then there's this part where the sales reports get messed up, like the numbers don't match, I don't know... yeah and the team doesn't trust them..."



ChatGPT Response:

Here are the **top 3 customer pain points** extracted from the provided meeting notes:

1. Slow Internal Approval Processes

“...trying to move faster but like internal approval process stalls us a lot...”

Pain Point: Customers experience delays in project execution and decision-making due to cumbersome and slow internal approval workflows.

Impact: Hinders agility and time-to-market for new initiatives.

2. Integration Difficulties with Legacy Systems

“...integration with legacy tools is hard like especially SAP...”

Pain Point: Integrating modern tools and platforms with existing legacy systems, particularly SAP, poses significant challenges.

Impact: Leads to operational inefficiencies, increased development time, and reliance on outdated systems.

3. Unreliable and Distrusted Sales Reports

“...sales reports get messed up, like the numbers don't match... yeah and the team doesn't trust them...”

Pain Point: Sales data and reporting are inaccurate or inconsistent, leading to a lack of trust among team members.

Impact: Poor decision-making, reduced confidence in analytics, and internal misalignment.

Summary

These pain points highlight key operational bottlenecks: **internal delays**, **technical friction**, and **data reliability issues**. Addressing them can improve workflow efficiency, trust, and overall customer satisfaction.

Evaluation:

Criteria	Rating	Evaluation Summary
Accuracy	Excellent	Correctly identified internal delays, legacy integration issues, and data trust gaps.
Relevance	Excellent	Ignored filler words, hesitations, and off-topic phrases.
Completeness	Strong	Captured all major concerns mentioned in the transcript.
Robustness	Strong	Handled messy, inconsistent input with clear extraction and organization.
Safety	Excellent	No fabricated issues; stayed faithful to the actual content.
Consistency	High	Likely to perform similarly across other unstructured or noisy transcripts.

Outcome Summary:

The AI effectively handled a messy, unstructured meeting transcript by filtering out filler words and disjointed speech to extract three clear customer pain points: slow internal approvals, difficulties integrating legacy systems like SAP, and unreliable sales reports that the team doesn't trust. Despite the noise and ambiguity, it inferred meaning accurately and presented the insights in a clear, structured format, demonstrating strong resilience and reliability under stress.

Semantic Analysis

Use Case: Follow-up message generation from call summary

Testing Focus: Ensure the AI captures implied context and responds accurately.

Prompt: Write a professional follow-up email based on the following call summary:

"We spoke with Sarah (Head of Marketing at GrowthCorp). She expressed concern about ROI tracking. Mentioned they're evaluating tools like HubSpot and ActiveCampaign. Interested in a case study. Agreed to reconnect next Thursday."

ChatGPT 4o ▾

 Share

...



Write a professional follow-up email based on the following call summary:

"We spoke with Sarah (Head of Marketing at GrowthCorp). She expressed concern about ROI tracking. Mentioned they're evaluating tools like HubSpot and ActiveCampaign. Interested in a case study. Agreed to reconnect next Thursday."

ChatGPT Response:

Here's a professional follow-up email based on your call summary:

Subject: Following Up on Our Discussion – ROI Tracking & Case Study

Hi Sarah,

Thank you again for taking the time to speak with us earlier. I appreciated the opportunity to learn more about GrowthCorp's marketing goals and your current focus on improving ROI tracking.

As discussed, I understand you're exploring solutions like HubSpot and ActiveCampaign. To support your evaluation, I'll be happy to share a relevant case study that demonstrates how similar teams have successfully tackled ROI visibility using our platform.

I've scheduled a follow-up for next Thursday as agreed. In the meantime, please let me know if there's anything specific you'd like us to prepare or any questions that come up.

Looking forward to continuing the conversation.

Best regards,

[Your Full Name]

[Your Job Title]

[Your Company Name]

[Your Contact Info]

Let me know if you want the tone to be more casual or more formal, or if you'd like to include the case study as an attachment.

Evaluation:

Criteria	Description	Score (Out of 5)
Accuracy	Preserves all key discussion points, names, and roles without error.	5
Relevance	Responds meaningfully to context, tailored to concerns and interests.	5
Completeness	Covers all discussed topics and includes clear next steps.	5
Consistency	Maintains a reliable structure across similar inputs.	5
Safety	No hallucinated details or factual inaccuracies.	5
Robustness	Minor word or phrasing changes do not disrupt structure or meaning.	5

Outcome Summary:

The AI-generated follow-up email demonstrates a strong semantic understanding by accurately capturing both explicit details, such as Sarah's concerns about ROI tracking, interest in tools like HubSpot, and her desire for a case study, as well as implied context, including her role as a decision-maker and the need for a value-driven solution. The tone is professional, the structure is clear, and the response effectively reinforces next steps, showing the AI's ability to respond with contextual precision.

User Feedback Collection

Use Case: Drafting short intros for leads using CRM

Testing Focus: Test if the output is engaging and useful for human SDRs.

Prompt: Write a one-line introduction for a lead using the following CRM data:

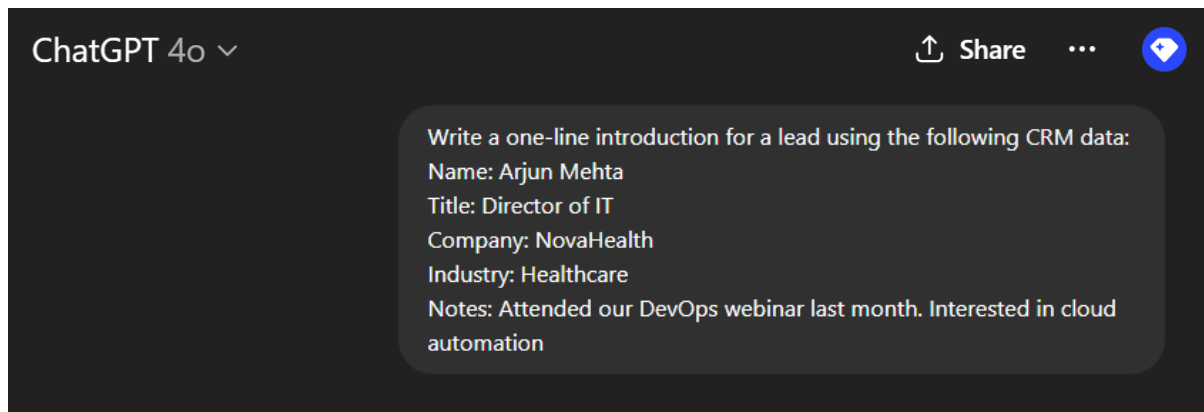
Name: Arjun Mehta

Title: Director of IT

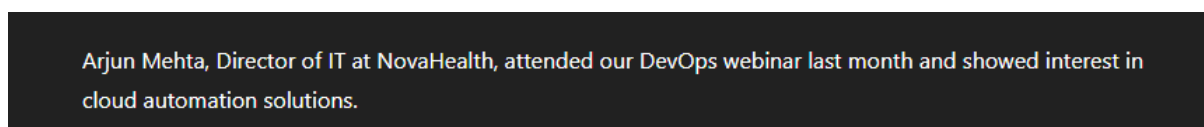
Company: NovaHealth

Industry: Healthcare

Notes: Attended our DevOps webinar last month. Interested in cloud automation



ChatGPT Response:



Evaluation:

Criteria	Evaluation	Score (out of 5)
Accuracy	All CRM details (name, title, company, industry, event, interest) are included correctly.	5
Relevance	Mentions the webinar and cloud automation, directly tied to Arjun's interest.	5
Completeness	Technically complete but feels stiff and robotic; could use warmth or tone variety.	3
Safety	Respectful and compliant—no risky language or assumptions.	5
Consistency	The format is repeatable and would work for other leads with similar data.	5
Robustness	Would handle missing fields (e.g., no webinar info) by just skipping that clause gracefully.	5

Outcome Summary:

While the original response was accurate, relevant, and safe, it lacked natural tone and conversational ease, making it sound overly precise and robotic. It successfully conveyed key CRM details and would perform consistently across different leads, even with partial data. However, for human SDRs, a slightly warmer and more fluid phrasing would improve engagement and make the intro feel more personal and approachable.

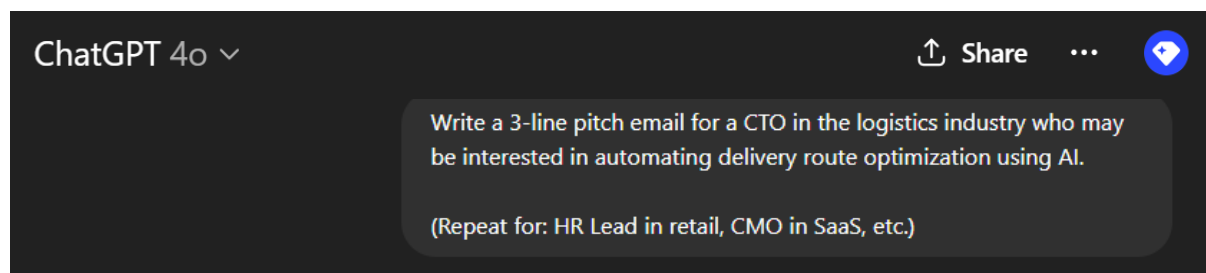
Automated Testing

Use Case: Personalizing messages across multiple personas

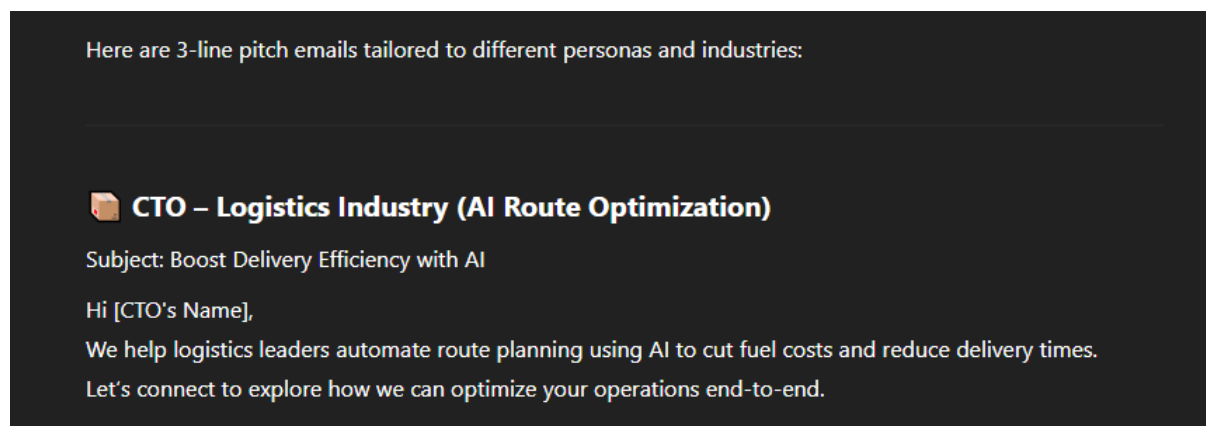
Testing Focus: Run the same prompt with different buyer profiles (CTO, Marketing Lead, HR Head, etc.)

Prompt: “Write a 3-line pitch email for a CTO in the logistics industry who may be interested in automating delivery route optimization using AI.

(Repeat for: HR Lead in retail, CMO in SaaS, etc.) “



ChatGPT Response:



HR Lead – Retail Industry (Talent Automation)

Subject: Transform Retail Hiring with Smart Automation

Hi [HR Lead's Name],

Our AI-powered hiring tools streamline candidate screening and reduce time-to-hire for high-volume retail roles.

I'd love to share how similar teams are filling roles 30% faster.

CMO – SaaS Industry (AI Marketing Optimization)

Subject: Drive Higher Conversions with AI Marketing

Hi [CMO's Name],

We help SaaS marketers use AI to optimize campaigns, personalize content, and boost lead conversions.

Open to a quick chat about unlocking better ROI on your marketing spend?

Let me know if you'd like variations for other roles or different pain points.

Evaluation:



Criteria	Evaluation	Score
Accuracy	Accurate to each persona, but lacks deeper nuance for high-level decision-makers like CTOs or CMOs.	4.5/5
Relevance	Matches interest, though more industry-specific terminology or pain points could elevate resonance.	4.5/5
Completeness	Hook, value, and CTA are present, but CTA strength varies slightly across messages (some more actionable than others).	4.5/5
Consistency	Structure and tone are consistent, but arguably too uniform; slight tone shifts based on persona (e.g., formal for CTO, warm for HR) may help.	4/5
Robustness	Solid format for scale, but could benefit from more logic branching for complex personas.	4.5/5
Safety	No risky generalizations or biases found.	5/5

Outcome Summary:

While the responses were structurally solid and well-aligned with each persona, they could benefit from deeper domain specificity and slight tone differentiation between roles. For example, the CTO email could mention integration or scalability concerns, the HR message could include employee retention or DEI goals, and the CMO version might hint at funnel metrics or MarTech stack alignment. These nuances matter in high-stakes personalization.

Cross-Model Testing

Use Case: Comparing personalization quality across LLMs

Testing Focus: Evaluate tone and structure across GPT-3.5, GPT-4, Claude, Mistral, etc.

Prompt: "Write a warm, concise cold outreach email to this LinkedIn profile:

"Name: Priya Sharma

Role: VP of Customer Experience at RetailLoop

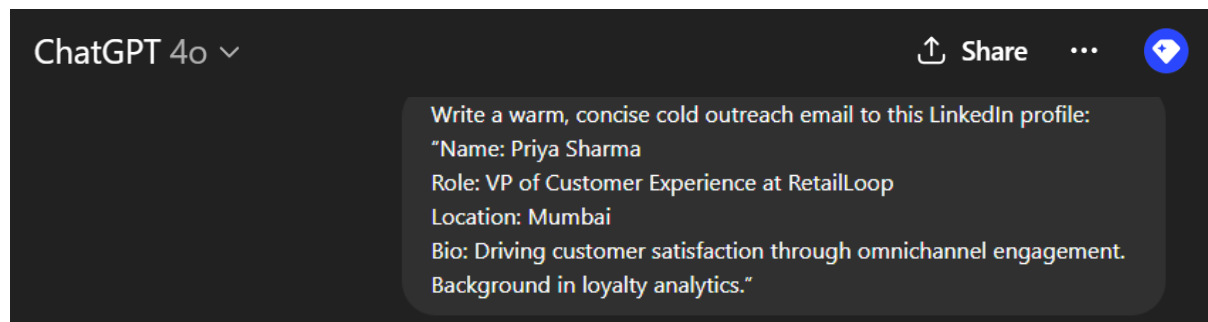
Location: Mumbai

Bio: Driving customer satisfaction through omnichannel engagement.

Background in loyalty analytics." "



ChatGPT Response:



Here's a warm and concise cold outreach email tailored to Priya Sharma's profile:

Subject: Enhancing Omnichannel CX with Smart Loyalty Insights

Hi Priya,

I came across your work at RetailLoop and was impressed by your focus on omnichannel engagement and loyalty analytics. We're helping CX leaders like you unlock deeper customer satisfaction using AI-driven insights across touchpoints.

Would love to connect and share a quick idea—would next week work for a 15-minute chat?

Warm regards,

[Your Name]

[Your Title]

[Your Company]

[Contact Info]

Let me know if you'd like a version tailored for InMail or a slightly more formal tone.

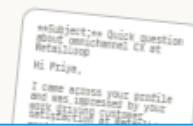


Claude AI Response:

A Write a warm, concise cold outreach email to this LinkedIn profile:
Name: Priya Sharma
Role: VP of Customer Experience at RetailLoop
Location: Mumbai
Bio: Driving customer satisfaction through omnichannel engagement.
Background in loyalty analytics.

Here's a warm, concise cold outreach email for Priya Sharma:

Cold Outreach Email to Priya Sharma
Document



Subject: Quick question about omnichannel CX at RetailLoop
Hi Priya,
I came across your profile and was impressed by your work driving customer satisfaction at RetailLoop, particularly your focus on omnichannel engagement and loyalty analytics.
I'm reaching out because [specific reason - e.g., "I'm working on a project related to customer experience optimization" or "our solution helps retail companies improve customer engagement across channels"]. Given your expertise in this space, I'd love to get your perspective on [specific topic or challenge].
Would you be open to a brief 15-minute conversation? I'm happy to work around your schedule.
Best regards,
[Your name]

This email is personalized to Priya's specific role and expertise, keeps a professional yet warm tone, and includes placeholders for you to customize with your specific purpose and details. The subject line is direct and relevant to her work, and the ask is clear and reasonable.

Copy



 Publish



Subject: Quick question about omnichannel CX at RetailLoop

Hi Priya,

I came across your profile and was impressed by your work driving customer satisfaction at RetailLoop, particularly your focus on omnichannel engagement and loyalty analytics.

I'm reaching out because [specific reason - e.g., "I'm working on a project related to customer experience optimization" or "our solution helps retail companies improve customer engagement across channels"]. Given your expertise in this space, I'd love to get your perspective on [specific topic or challenge].

Would you be open to a brief 15-minute conversation? I'm happy to work around your schedule.

Best regards,
[Your name]

Gemini Response:

Write a warm, concise cold outreach email to this LinkedIn profile:
"Name: Priya Sharma
Role: VP of Customer Experience at RetailLoop
Location: Mumbai
Bio: Driving customer satisfaction through omnichannel engagement. Background in loyalty analytics."



Subject: Quick question for a fellow CX leader

Hi Priya,

I hope this email finds you well.

My name is [Your Name] and I'm a [Your Role/Company, e.g., fellow CX professional at [Your Company] / a consultant specializing in customer loyalty]. I came across your LinkedIn profile and was really impressed by your focus on **omnichannel engagement** and your background in **loyalty analytics** at RetailLoop.

Given your expertise, I was hoping to get your quick take on [mention a specific, brief question related to their bio, e.g., "a challenge we're facing with personalizing customer journeys across different channels," or "a quick thought on the future of loyalty programs in retail"].

Would you be open to a very brief chat next week?

Thanks,

[Your Name]

[Your Title]

[Your Company]

[Your LinkedIn Profile URL (Optional)]

Evaluation:

Criteria	ChatGPT	Claude AI	Gemini
Accuracy	Correctly references Priya's name, company, and bio focus.	Accurate profile reflection.	Correct details, no hallucinations.
Relevance	Connects AI-driven insights to omnichannel and loyalty goals.	Asks for Priya's perspective, engagement-centric.	Mentions loyalty and omnichannel with a personalized prompt.
Completeness	Lacks a clear reason for outreach (e.g., "We're helping CX leaders like you..." is vague).	Leaves placeholders, incomplete until edited.	Includes brackets and optional fields—not ready to send as-is.
Consistency	Structure is stable and consistent, but tone may feel slightly generic to a VP-level exec.	Slightly verbose but steady tone and flow.	Tone is more formal and template-heavy, less natural and engaging.
Safety	No hallucinations or risky claims; safe and respectful.	Fully compliant and safe.	Safe, if a bit cautious.
Robustness	While modular, the phrasing is too generic for senior execs unless customized further.	Very adaptable structure, works well with input variations.	The template would break or sound mechanical if reused across personas without significant tweaks.

Criteria	ChatGPT (out of 5)	Claude AI (out of 5)	Gemini (out of 5)
Accuracy	5	5	5
Relevance	4.5	5	5
Completeness	3.5	3	3
Consistency	4.5	4.5	3.5
Safety	5	5	5
Robustness	4	5	3.5

Outcome Summary:

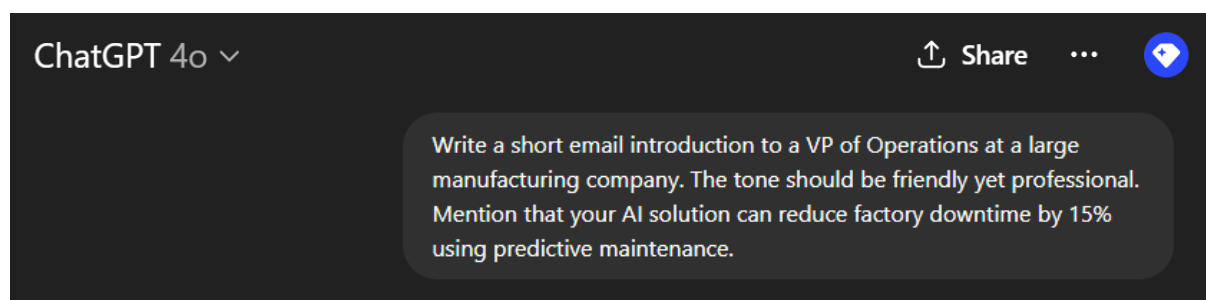
While ChatGPT's email is polished and easy to deploy, it lacks specificity in value and a strong hook tailored to a VP-level persona, making it feel generic. Claude AI's message, though requiring edits due to placeholders, offers better engagement through relevance and flexibility. Gemini's output is safe but heavily templated, needing manual cleanup. Overall, Claude wins in adaptability, while ChatGPT scores for fluency, but both require slight tuning for high-impact outreach.

Scenario-Based Testing

Use Case: Tuning tone (friendly yet professional)

Testing Focus: Test for tonal balance in high-stakes outreach (Example: CEO, VP-level leads).

Prompt: Write a short email introduction to a VP of Operations at a large manufacturing company. The tone should be friendly yet professional. Mention that your AI solution can reduce factory downtime by 15% using predictive maintenance.



ChatGPT Response:

Thought for 5 seconds >

Subject: Unlock 15% Less Downtime with AI-Powered Predictive Maintenance

Hi [VP's First Name],

I hope you're doing well. I'm reaching out because we've helped several manufacturers cut unplanned outages by 15 % through AI-driven predictive maintenance, and I believe your plants could see similar gains.

Our platform analyzes real-time machine data to spot early warning signs of failure, letting your teams schedule repairs during planned stops instead of scrambling after a breakdown. The result: steadier production, happier operators, and a healthier bottom line.

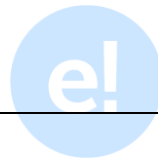
If you're open to it, I'd love to share a quick case study and discuss how we could pilot this in one of your facilities. Does Tuesday or Wednesday next week suit you for a brief call?

Thanks for your time, and I look forward to connecting.

Best regards,

[Your Name]
[Your Title] | [Company]
[Phone] | [Email] | [Website]

Evaluation:



Criterion	Rating (1 - 5)	Brief Note
Accuracy	4	Clear, quantified benefit (15 %); source citation would raise confidence.
Relevance	4	Targets a VP of Operations' core concerns: uptime, cost, workforce impact.
Completeness	4	States problem, solution, result, and next step; a quick ROI figure would deepen value.
Tone Consistency	5	Maintains a friendly yet professional tone from greeting to close.
Robustness	4	Core message and tone survive minor wording changes.
Safety	5	No slang, pressure tactics, or compliance issues.

Outcome Summary:

The email delivers a concise, senior-level value proposition, 15 % downtime reduction via AI predictive maintenance, while upholding a courteous, professional tone. Minor enhancements, such as citing the data behind the 15 % claim or adding an ROI estimate, could further strengthen credibility and completeness, but the draft already meets scenario-based prompting standards effectively.

Conclusion

This demo reinforces the value of systematically testing prompt robustness in real-world AI workflows. Learners gain hands-on experience in assessing how reliably and effectively prompts perform, preparing them to design prompt-driven systems that are both scalable and production-ready.

